Solutions: Additional Exercises 4

Gradient, Directional Derivatives & Higher-Order Partial Derivatives

Exercise 1: Directional derivatives

(a) $f = x^2 + xy - y^2$. $\nabla f = (2x + y, x - 2y)$. At $(1, 2)$: $\nabla f = (4, -3)$. Unit vector: $\hat{u} = (3, 4)/5 = (3/5, 4/5)$. $D_{\hat{u}} f = (4)(3/5) + (-3)(4/5) = 12/5 - 12/5 = 0$.

(b) $f = xy + yz + zx$. $\nabla f = (y + z, x + z, y + x)$. At $(2, 1, 3)$: $\nabla f = (4, 5, 3)$. $|\bar{a}| = \sqrt{9 + 16 + 144} = 13$. $\hat{a} = (3, 4, 12)/13$. $D_{\hat{a}} f = (4 \cdot 3 + 5 \cdot 4 + 3 \cdot 12)/13 = (12 + 20 + 36)/13 = 68/13$.

(c) $f = e^x \cos y$. $\nabla f = (e^x \cos y, -e^x \sin y)$. At $(0, 0)$: $\nabla f = (1, 0)$. $\hat{u} = (\cos(\pi/4), \sin(\pi/4)) = (1/\sqrt{2}, 1/\sqrt{2})$. $D_{\hat{u}} f = 1/\sqrt{2}$.

(d) $f = \ln(x^2 + y^2 + z^2)$. $\nabla f = \frac{2(x,y,z)}{x^2+y^2+z^2}$. At $(1, 2, 1)$: $r^2 = 6$, $\nabla f = (2/6, 4/6, 2/6) = (1/3, 2/3, 1/3)$. $|\bar{a}| = \sqrt{4 + 16 + 16} = 6$. $\hat{a} = (2, 4, 4)/6 = (1/3, 2/3, 2/3)$. $D_{\hat{a}} f = (1/3)(1/3) + (2/3)(2/3) + (1/3)(2/3) = 1/9 + 4/9 + 2/9 = 7/9$.

(e) $f = xyz$. $\nabla f = (yz, xz, xy)$. At $(2, 4, 2)$: $\nabla f = (8, 4, 8)$. $|\bar{u}| = \sqrt{16 + 4 + 16} = 6$. $\hat{u} = (4, 2, -4)/6 = (2/3, 1/3, -2/3)$. $D_{\hat{u}} f = 8(2/3) + 4(1/3) + 8(-2/3) = 16/3 + 4/3 - 16/3 = 4/3$.

(f) $f = x^2 y$. $\nabla f = (2xy, x^2)$. At $(1, 1)$: $\nabla f = (2, 1)$. $\hat{u} = (\cos(\pi/3), \sin(\pi/3)) = (1/2, \sqrt{3}/2)$. $D_{\hat{u}} f = 2(1/2) + 1(\sqrt{3}/2) = 1 + \sqrt{3}/2$.

(g) $f(x, y) = x^4 y^2/(x^4 + y^2)$ at $(0, 0)$, $\bar{u} = (1, 1)$, $\hat{u} = (1/\sqrt{2}, 1/\sqrt{2})$. $D_{\hat{u}} f(0, 0) = \lim_{t\to 0} \frac{f(t/\sqrt{2},t/\sqrt{2})-0}{t}$. $f(t/\sqrt{2}, t/\sqrt{2}) = \frac{(t/\sqrt{2})^4 (t/\sqrt{2})^2}{(t/\sqrt{2})^4+(t/\sqrt{2})^2} = \frac{t^6/8}{t^4/4+t^2/2} = \frac{t^6/8}{t^2(t^2/4+1/2)} = \frac{t^4/8}{t^2/4+1/2}$. As $t \to 0$: $\frac{0}{1/2} = 0$. So $D_{\hat{u}} f(0, 0) = \lim_{t\to 0} \frac{0}{t} = 0$. **Answer:** 0.

Exercise 2: Finding directions

(a) $f = 3x + 4y$. $\nabla f = (3, 4)$ (constant). Direction of steepest ascent: $\hat{u} = (3, 4)/5$. Rate (maximum directional derivative): $|\nabla f| = \sqrt{9 + 16} = 5$.

(b) $f = x^2 y + y^3$. $\nabla f = (2xy, x^2 + 3y^2)$. At $(1, -1)$: $\nabla f = (-2, 4)$. Want $D_{\hat{u}} f = \nabla f \cdot \hat{u} = 0$: $\hat{u} \perp (-2, 4)$, so $\hat{u} = \pm(4, 2)/\sqrt{20} = \pm(2, 1)/\sqrt{5}$.

(c) $f = xy + z$. $\nabla f = (y, x, 1)$. At $(0, 1, -2)$: $\nabla f = (1, 0, 1)$, $|\nabla f| = \sqrt{2}$. $D_{\hat{u}} f = \nabla f \cdot \hat{u}$. Maximum is $|\nabla f| = \sqrt{2} < 1$? No: $\sqrt{2} \approx 1.41 > 1$. $D_{\hat{u}} f = 1$ is achievable: need $\nabla f \cdot \hat{u} = 1$, i.e. $\hat{u}_1 + \hat{u}_3 = 1$. Take $\hat{u} = (1/\sqrt{2}, 0, 1/\sqrt{2})$... check: $(1)(1/\sqrt{2}) + (0)(0) + (1)(1/\sqrt{2}) = 2/\sqrt{2} = \sqrt{2} \neq 1$. Hmm. Solve: $(1, 0, 1) \cdot \hat{u} = 1$, $|\hat{u}| = 1$.

Let $\hat{u} = (a,b,c)$: $a + c = 1$, $a^2 + b^2 + c^2 = 1$. One solution: $b = 0$, $c = 1 - a$, $a^2 + (1-a)^2 = 1 \Rightarrow 2a^2 - 2a = 0 \Rightarrow a = 0$ or $a = 1$. So $\hat{u} = (0,0,1)$ or $\hat{u} = (1,0,0)$.

For $D_{\hat{u}}f = 20$: need $\nabla f \cdot \hat{u} = 20$, but $|\nabla f \cdot \hat{u}| \le |\nabla f| = \sqrt{2} < 20$. **No such direction exists.**

(d) $f = x^2y + y^3$. $|\nabla f(1,-1)| = |(-2,4)| = \sqrt{20} = 2\sqrt{5} \approx 4.47 < 20$. **No such direction exists**, since $|D_{\hat{u}}f| \le |\nabla f| = 2\sqrt{5} < 20$.

Exercise 3: Gradient at a point

(a) $u = x + y + 2xy$. $u_x = 1 + 2y$, $u_y = 1 + 2x$. At $(1,1)$: $\nabla u = (3,3)$.

(b) $u = x^2 + y^2 + z^2$. $\nabla u = (2x, 2y, 2z)$. At $(2,0,0)$: $\nabla u = (4,0,0)$.

(c) $u = \ln(x^2 + y^2 + z^2)$. $\nabla u = \frac{2(x,y,z)}{x^2+y^2+z^2}$. At $(1,1,1)$: $r^2 = 3$, $\nabla u = (2/3, 2/3, 2/3)$.

(d) $u = x^2e^{y-z}$. $u_x = 2xe^{y-z}$, $u_y = x^2e^{y-z}$, $u_z = -x^2e^{y-z}$. At $(2,1,1)$: $e^0 = 1$. $\nabla u = (4,4,-4)$.

(e) $u = \arctan(y/x) + \arctan(x/y)$. Both are functions of y/x and $x/y = 1/(y/x)$. $\arctan(t) + \arctan(1/t) = \pi/2$ for $t > 0$, $-\pi/2$ for $t < 0$ (constant on each quadrant). So u is constant (piecewise), $\nabla u = \mathbf{0}$.

(f) $u = \arctan(xyz) + \operatorname{arccot}(xyz)$. Since $\arctan(t) + \operatorname{arccot}(t) = \pi/2$ for all t, $u = \pi/2$ is constant, so $\nabla u = \mathbf{0}$.

(g) $u = f(r)$, $r = \sqrt{x^2 + y^2 + z^2}$. $u_x = f'(r) \cdot \frac{x}{r}$. $\nabla u = f'(r) \cdot \frac{(x,y,z)}{r} = \frac{f'(r)}{r}\bar{r}$.

Exercise 4: Second-order partial derivatives

(a) $z = \sin(xy)$. $z_x = y\cos(xy)$. $z_{xy} = \cos(xy) - xy\sin(xy)$. $z_{xyy} = -x\sin(xy) \cdot 1 + (-x\sin(xy) - x^2y\cos(xy))$... Let's compute systematically: $\frac{\partial^2 z}{\partial x\,\partial y^2} = \frac{\partial}{\partial x}(z_{yy})$. $z_y = x\cos(xy)$. $z_{yy} = -x^2\sin(xy)$. $z_{x(yy)} = -2x\sin(xy) - x^2y\cos(xy)$.

(b) $z = x^3y + xy^3$. $z_x = 3x^2y + y^3$, $z_{xy} = 3x^2 + 3y^2$. $z_y = x^3 + 3xy^2$, $z_{yx} = 3x^2 + 3y^2$. $f_{xy} = f_{yx} = 3x^2 + 3y^2$. ✓

(c) $z = x^y = e^{y\ln x}$. $z_x = yx^{y-1}$, $z_{xy} = x^{y-1} + yx^{y-1}\ln x = x^{y-1}(1 + y\ln x)$. $z_y = x^y\ln x$, $z_{yx} = yx^{y-1}\ln x + x^y/x = x^{y-1}(y\ln x + 1)$. Equal. ✓

(d) $u = \arctan(y/x)$. $u_x = -y/(x^2+y^2)$. $u_{xx} = \frac{2xy}{(x^2+y^2)^2}$. $u_y = x/(x^2+y^2)$. $u_{yy} = \frac{-2xy}{(x^2+y^2)^2}$. $u_{xx} + u_{yy} = 0$. ✓

(e) $u = e^x(x\cos y - y\sin y)$. $u_x = e^x(x\cos y - y\sin y) + e^x\cos y = e^x((x+1)\cos y - y\sin y)$. $u_{xx} = e^x((x+2)\cos y - y\sin y)$. $u_y = e^x(-x\sin y - \sin y - y\cos y) = e^x(-(x+1)\sin y - y\cos y)$. $u_{yy} = e^x(-(x+1)\cos y - \cos y + y\sin y) = e^x(-(x+2)\cos y + y\sin y)$. $u_{xx} + u_{yy} = e^x[(x+2)\cos y - y\sin y - (x+2)\cos y + y\sin y] = 0$. ✓

(f) $u = \ln(1/r) = -\ln r$, $r = \sqrt{(x-a)^2 + (y-b)^2}$. $u_x = -r_x/r = -(x-a)/r^2$. $u_{xx} = -r^2 - (x-a)2r \cdot r_x)/r^4$... more carefully: $u_{xx} = \frac{-(r^2-(x-a)^2 \cdot 2)/r^2}{r^2} \dots$
Let $X = x-a$, $Y = y-b$, $r^2 = X^2+Y^2$. $u = -\frac{1}{2}\ln(X^2+Y^2)$. $u_X = -X/(X^2+Y^2)$. $u_{XX} = \frac{-(X^2+Y^2)+2X^2}{(X^2+Y^2)^2} = \frac{Y^2-X^2}{(X^2+Y^2)^2}$. $u_{YY} = \frac{X^2-Y^2}{(X^2+Y^2)^2}$. $u_{XX} + u_{YY} = 0$. ✓

Exercise 5: Second-order differential

(a) $u = x^2y^3$. $u_{xx} = 2y^3$, $u_{xy} = 6xy^2$, $u_{yy} = 6x^2y$. $d^2u = 2y^3(dx)^2 + 12xy^2\,dx\,dy + 6x^2y(dy)^2$.

(b) $u = e^{xy}$. $u_{xx} = y^2e^{xy}$, $u_{xy} = e^{xy} + xye^{xy} = (1+xy)e^{xy}$, $u_{yy} = x^2e^{xy}$. $d^2u = y^2e^{xy}(dx)^2 + 2(1+xy)e^{xy}\,dx\,dy + x^2e^{xy}(dy)^2$.

(c) $u = x^y$. $u_{xx} = y(y-1)x^{y-2}$, $u_{xy} = x^{y-1}(1+y\ln x)$, $u_{yy} = x^y(\ln x)^2$. $d^2u = y(y-1)x^{y-2}(dx)^2 + 2x^{y-1}(1+y\ln x)\,dx\,dy + x^y(\ln x)^2(dy)^2$.

(d) $u = x\ln(xy) = x\ln x + x\ln y$. $u_x = \ln x + 1 + \ln y = \ln(xy)+1$. $u_y = x/y$. $u_{xx} = 1/x$. $u_{xy} = 1/y$. $u_{yy} = -x/y^2$. $d^2u = \frac{(dx)^2}{x} + \frac{2\,dx\,dy}{y} - \frac{x(dy)^2}{y^2}$.

(e) $u = \sin(x+y) + \cos(x-y)$. $u_x = \cos(x+y) - \sin(x-y)$. $u_y = \cos(x+y) + \sin(x-y)$. $u_{xx} = -\sin(x+y) - \cos(x-y)$. $u_{xy} = -\sin(x+y) + \cos(x-y)$. $u_{yy} = -\sin(x+y) - \cos(x-y)$. $d^2u = [-\sin(x+y) - \cos(x-y)](dx)^2 + 2[-\sin(x+y) + \cos(x-y)]\,dx\,dy + [-\sin(x+y) - \cos(x-y)](dy)^2$.

(f) $u = f(t)$ where $t = x^2 - y^2$. $u_x = f'(t) \cdot 2x$. $u_{xx} = f''(t) \cdot 4x^2 + 2f'(t)$. $u_y = f'(t) \cdot (-2y)$. $u_{yy} = f''(t) \cdot 4y^2 - 2f'(t)$. $u_{xy} = f''(t) \cdot 2x \cdot (-2y) = -4xyf''(t)$. $d^2u = (4x^2f'' + 2f')(dx)^2 - 8xyf''\,dx\,dy + (4y^2f'' - 2f')(dy)^2$.

(g) $u = \frac{1}{2}\ln(x^2+y^2)$. $u_x = x/(x^2+y^2)$, $u_{xx} = (y^2-x^2)/(x^2+y^2)^2$, $u_{yy} = (x^2-y^2)/(x^2+y^2)^2$. $u_{xx} + u_{yy} = 0$. ✓ $u_{xy} = -2xy/(x^2+y^2)^2$. $d^2u = \frac{y^2-x^2}{(x^2+y^2)^2}(dx)^2 - \frac{4xy}{(x^2+y^2)^2}\,dx\,dy + \frac{x^2-y^2}{(x^2+y^2)^2}(dy)^2$.